

Source A

Alpha particles.

Properties - It is +vely charged

- low penetrating power
- short range in air.
- High ionising power
- Stopped by thin sheet of paper
- Deflected ~~toward~~ by both Electric / Magnetic field.

Source B - Beta particle

Properties - Negatively charged

- Higher penetrating power than α -ples
- Higher range in air than α -ples.
- lower ionising power than α -ples
- fast moving electron
- stopped by thin aluminium foil.
- Deflected by both Electric / Magnetic field.

Source C - Gamma ray.

Properties - No charge

- Not deflected in both electric / Magnetic field
- Highest penetrating power
- shortest range in air.
- Lowest ionising power.
- Travel faster at as speed of light.

c) 234g of Thorium contain 6.02×10^{23} molecules.

$$2g \text{ of Thorium contain } \frac{2 \times 6.02 \times 10^{23}}{234}$$

$$= 5.1453 \times 10^{21} \text{ Molecules.}$$

$$\text{Decay constant } \lambda = \frac{0.693}{t_{1/2}} \left(\frac{\ln 2}{t_{1/2}} \right)$$

$$= \frac{0.693}{5.45 \times 365 \times 24 \times 60 \times 60}$$

$$= 4.032 \times 10^{-9} \text{ s}^{-1}$$

$$\text{Activity, } A = \lambda N$$

$$= 4.032 \times 10^{-9} \times 5.1453 \times 10^{21}$$

$$= 2.0746 \times 10^{13} \text{ Bq (s}^{-1}\text{)}$$

$$[LHS] = [RHS] \quad \text{ITEM 2}$$

$$(a) \quad P \propto L^x v^y \rho^z$$

$$P = k L^x v^y \rho^z$$

$$[P] = [k] [L]^x [v]^y [\rho]^z$$

$$ML^2T^{-3} = L^x (LT^{-1})^y (ML^{-3})^z$$

$$ML^2T^{-3} = M^z L^{x+y-3z} T^{-y}$$

M

$$2 = 1 \quad \text{--- (1)}$$

T

$$-3 = -y$$

$$y = 3 \quad \text{--- (2)}$$

L

$$2 = x + y - 3z$$

$$2 = x + 3 - 3(1)$$

$$x = 2$$

$$\therefore P = k L^2 v^3 \rho$$

$$(b) \quad [LHS] = [RHS]$$

$$F = \frac{1}{2} C \rho A v^2$$

$$[F] = \left[\frac{1}{2}\right] [C] [\rho] [A] [v]^2$$

$$MLT^{-2} = [C] ML^{-3} L^2 (LT^{-1})^2$$

$$MLT^{-2} = [C] MLT^{-2}$$

$$[C] = \frac{MLT^{-2}}{MLT^{-2}}$$

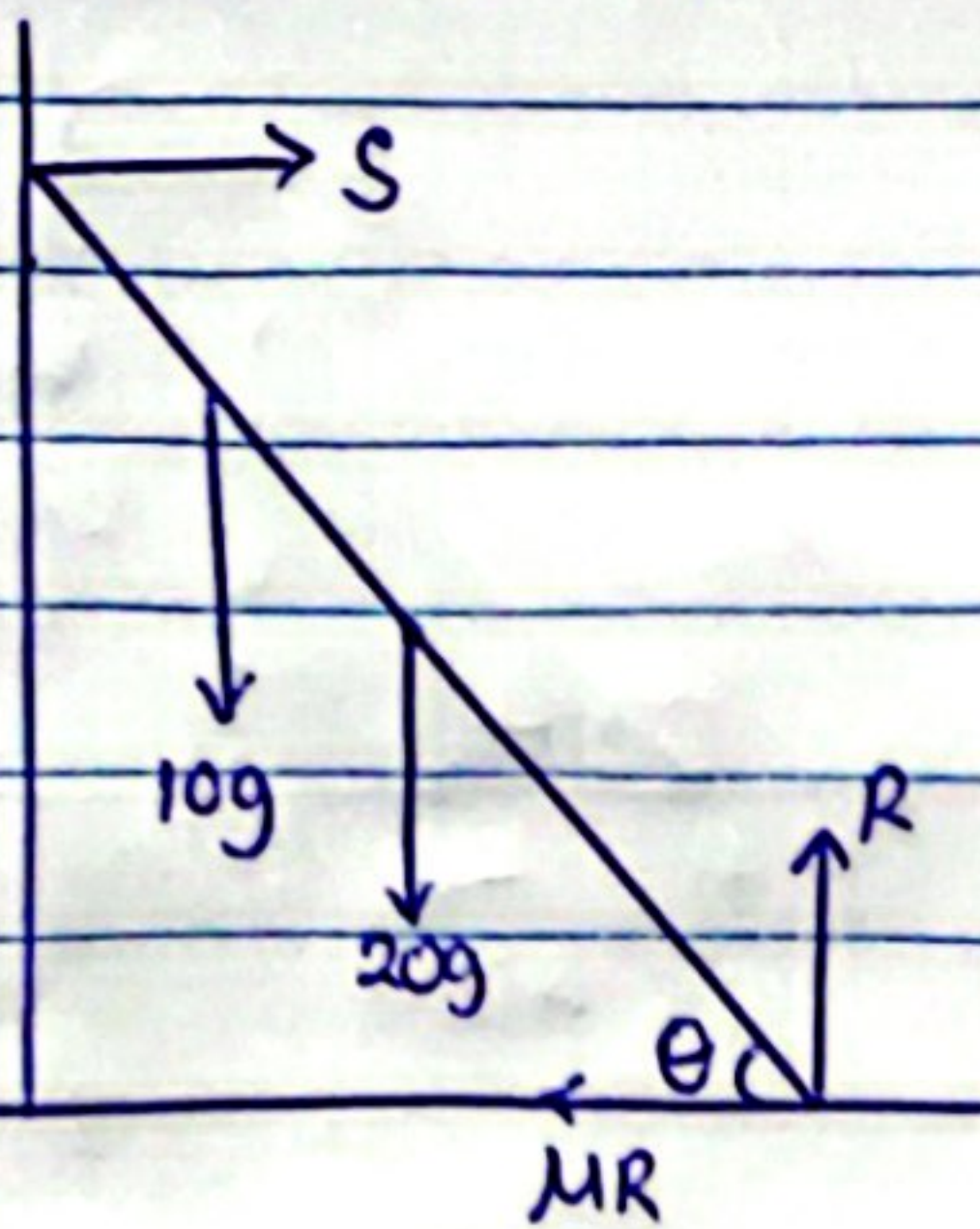
$$= 1$$

Since $[C] = 1$, then C is a constant and it is dimensionless

(c) For a body in mechanical equilibrium, the sum of clockwise moments about any point is equal to the sum of clockwise moments about the same point.

ITEM 2

(c)



Vertically

Vector sum of all forces in any direction is zero

Vertically

$$R = 10g + 20g$$

$$= 30g \quad \text{--- (1)}$$

Horizontally

$$S = MR$$

$$S = 0.5R \quad \text{--- (2)}$$

From (1) and (2);

$$S = 0.5 \times 30g$$

$$= 147.15$$

Sum of ^{clockwise} moments about any point is equal to the sum of anticlockwise moments about the same point.

Taking moments about the contact between ladder and ground;

$$10g \times \frac{3}{4} \cos \theta + 20g \times 0.5 \cos \theta = 147.15 \sin \theta$$

$$171.675 \cos \theta = 147.15 \sin \theta$$

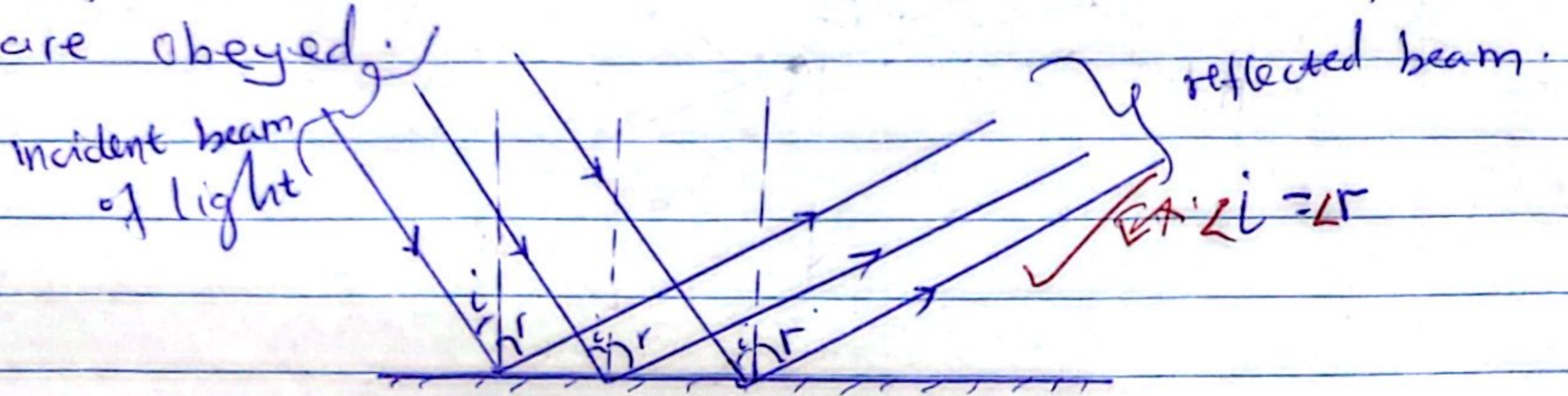
$$\tan \theta = \frac{171.675}{147.15}$$

$$\theta = 49.40^\circ$$

$\therefore$ The ladder makes an angle of 49.40° with the ground.

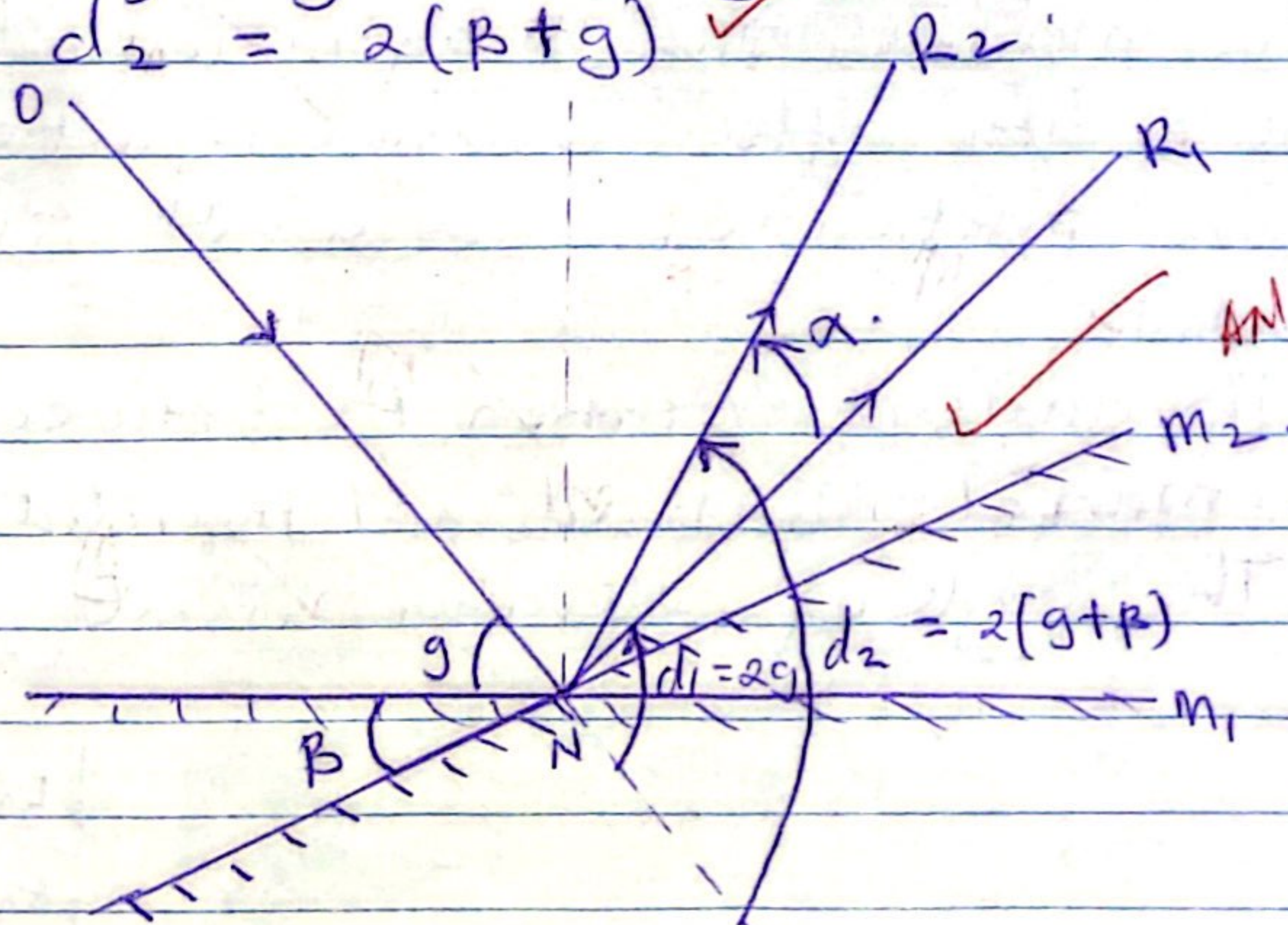
- a) Incident ray, reflected ray and the normal all lie in the principal axis.
The angle of reflection is equal to the angle of refraction.

b) Regular reflection. It takes place on smooth shiny surfaces like mirrors, water (clear). All laws of reflection are obeyed.



c) Consider a beam of light (constant) incident on a plane mirror with a glancing angle g , the deviation caused $d_1 = 2g$.

Let the angle of rotation of the mirror be β . When the mirror is rotated through an angle β , the new glancing angle is $\beta + g$, the new deviation caused $d_2 = 2(\beta + g)$.



$$\angle R_1 N R_2 = \angle R_2 N D - \angle R_1 N D$$

$$\alpha = 2g + 2\beta - 2g$$

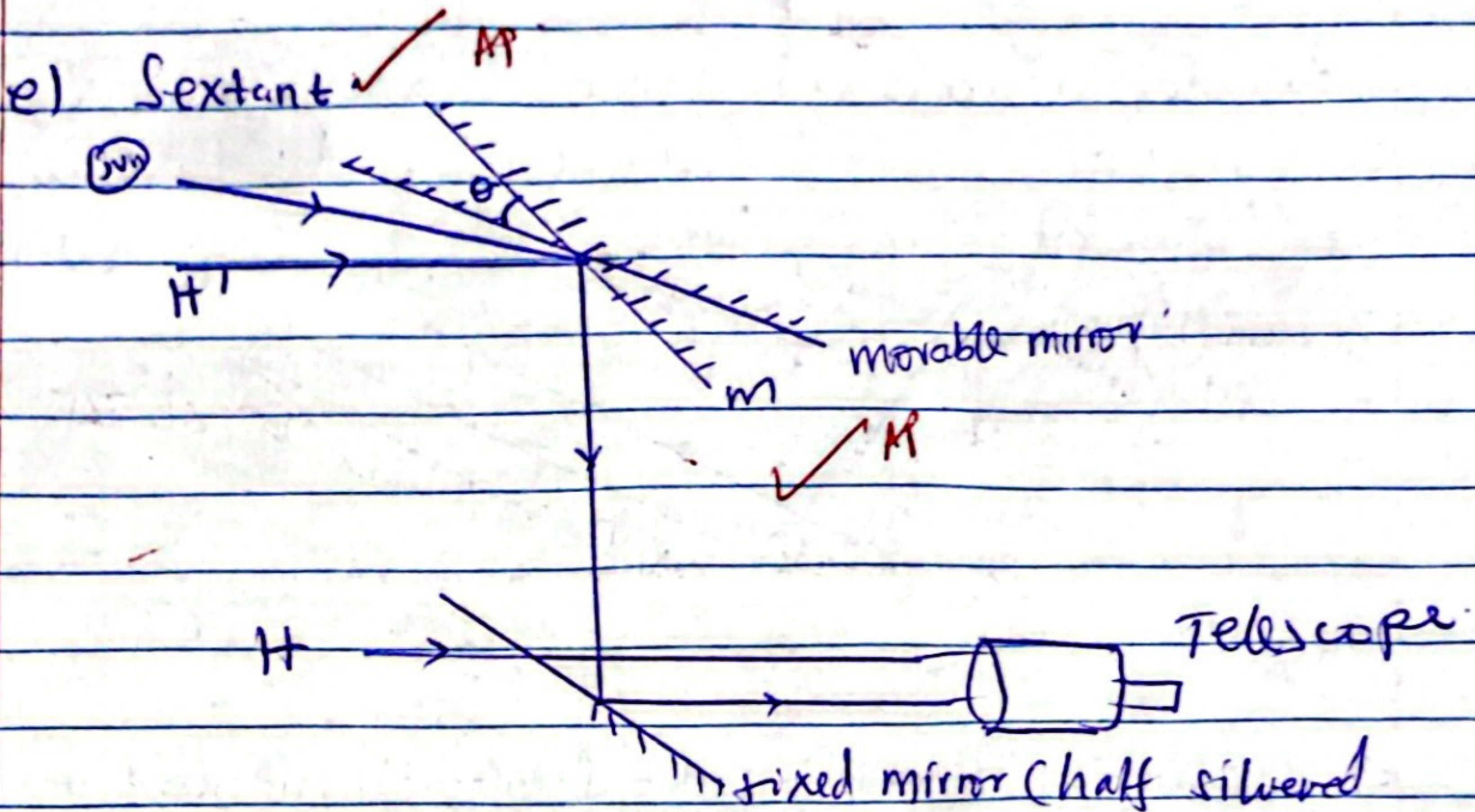
$\alpha = 2\beta$ (Hence the reflected ray is rotated twice the angle of rotation of the mirror)

d) $\alpha = 2\beta$

$\alpha = 2 \times 20$

$\alpha = 40^\circ$

Therefore the reflected ray will be rotated through 40° .



Horizon H is viewed directly through the unsilvered part of the mirror by means of a telescope.

The movable mirror is rotated until the image of horizon H' after successive reflection coincides with horizon H .

The position of the movable mirror is marked.

The movable mirror is again rotated until the image of the sun/star after successive reflection coincides with H .

The position of the movable mirror is again marked.

The angle of rotation θ between the two points is marked, measured and recorded.

The angle of elevation is 2θ .

Ex	05
AN	07
AP	08
Total score	20